

Studox.ai Career Assessment Roadmap Flow

Manual roadmap generation logic based on the 8-question career assignment. This document records the agreed product flow before implementation and future AI integration.

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1. Core Product Decision

- For now, Studox.ai will generate roadmaps manually through predefined logic, not fully dependent on an AI API key.
- Later, when API key, scale, and more users are ready, AI can replace or enhance the manual generation layer.
- The roadmap must be simple, direct, and based on the user's selected career assignment answers.
- If the user selects Full Stack Developer, the roadmap must only be Full Stack related. It should not include AI/ML headings or unrelated content.

2. Career Assignment Questions

No.	Question	Options / Type	Rule
1	Career goal	Full Stack Developer, AI/ML Engineer, Data Analyst, Cybersecurity, UX/UI Designer, Web Development, Other	Required. Controls roadmap domain and first portion of the roadmap.
2	Current level	Beginner, Basic coding knowledge, Intermediate, Advanced	Required. Controls roadmap level and number of options.
3	Timeline	1 month, 3 months, 6 months, 12 months	Required. Controls duration; recommendation depends on level.
4	Weekly time	3-5 hours, 6-8 hours, 9-12 hours, 15+ hours	Required. Controls workload and finish estimate.
5	Main focus	Job-ready skills, Internship preparation, Portfolio project, Optimal for coding interview preparation	Optional. Refines goal and interview preparation.
6	Projects built	0 projects, 1-2 projects, 3-5 projects, 5+ projects	Optional. Refines project difficulty.
7	Learning style	Video lessons, Practice tasks, Projects, Reading notes, Optimal learning	Optional. Refines learning format.
8	Extra context	Textarea	Optional. Adds personal context if provided.

Important UI updates agreed: remove Other from all radio questions, remove Not sure yet from Career goal, add Web Development, and change button text from Generate 3 Roadmaps to Generate Roadmap.

3. Required vs Optional Logic

Only the first four questions are mandatory. The remaining questions should never block roadmap generation.

Q1 + Q2 + Q3 + Q4 = proper roadmap generation. Q5-Q8 = optional personalization.

- If optional questions are answered, the roadmap becomes more personalized.
- If optional questions are not answered, default values should be used silently.
- The user should not be forced to answer optional questions before generating a roadmap.

4. Q1 Career Goal Logic

- Career goal decides the roadmap domain and first portion of the roadmap.

- Full Stack Developer should produce Full Stack content: HTML/CSS/JavaScript, React, Node.js, MongoDB, APIs, deployment, portfolio.
- AI/ML Engineer should produce AI/ML content: Python, math basics, data preprocessing, ML foundations, model building, AI projects.
- Data Analyst should produce Data Analyst content: Excel, SQL, Python for data, statistics, dashboards, BI tools.
- Cybersecurity should produce security content: networking, Linux, web security, ethical hacking basics, security projects.
- UI/UX Designer should produce design content: design fundamentals, Figma, user research, wireframes, portfolio case studies.
- Web Development should produce web-focused content: HTML/CSS/JavaScript, responsive UI, frontend basics, web projects.

5. Q2 Level Logic

Selected Level	Roadmap Options Shown	Selection Rule
Beginner	1 roadmap: Beginner	Only beginner roadmap is shown. No intermediate or advanced roadmaps are shown.
Basic coding knowledge	Beginner optional, Intermediate recommended, Advanced locked	Beginner selectable. Intermediate selectable. Advanced visible but locked.
Intermediate	Beginner optional, Intermediate recommended, Advanced locked	Beginner selectable. Intermediate selectable. Advanced visible but locked.
Advanced	1 roadmap: Advanced	Only advanced roadmap is shown. No beginner/intermediate roadmaps are shown.

Locked advanced card message suggestion: Advanced roadmap unlocks after completing intermediate milestones.

6. Q3 Timeline Logic

Timeline recommendations should depend on career goal and selected level. The same timeline options remain visible, but the best option should be highlighted as recommended.

Career Goal	Beginner / Basic	Intermediate	Advanced
Full Stack Developer	6 months	3-6 months	1-3 months
AI/ML Engineer	6-12 months	3-6 months	3 months
Data Analyst	3-6 months	1-3 months	1 month
Cybersecurity	6-12 months	3-6 months	3 months
UI/UX Designer	3-6 months	1-3 months	1 month
Web Development	3-6 months	1-3 months	1 month

- If user chooses an aggressive timeline, roadmap can still generate but should be treated as intensive.
- Example: Beginner + AI/ML + 1 month means high daily effort and limited depth.

7. Q4 Weekly Time Logic

Weekly Time	Roadmap Load	Expected Behavior
3-5 hours	Light	Fewer tasks per week, longer estimated finish time, more revision buffer.
6-8 hours	Balanced	Normal weekly plan, realistic project and practice load.
9-12 hours	Focused	More practice and project work, faster completion.
15+ hours	Intensive	Fastest path, higher weekly workload, less buffer.

Roadmap cards should show estimated completion such as: Estimated completion: 18 weeks at 9-12 hrs/week.

8. Manual Generation Flow

The current implementation should be manual first. The generation logic should select from predefined roadmap templates based on Q1-Q4 and then apply optional refinements from Q5-Q8.

Career Goal -> Level Scenario -> Timeline Recommendation -> Weekly Load -> Roadmap Cards

- Step 1: Read career goal and select matching domain template.
- Step 2: Read current level and decide whether to show 1 roadmap or multiple cards with locked state.
- Step 3: Read timeline and highlight recommended duration for that goal + level.
- Step 4: Read weekly time and calculate workload and estimated finish.
- Step 5: Apply optional focus, project experience, learning style, and extra context if provided.

9. Roadmap Card Output Rules

- Button text should be Generate Roadmap, not Generate 3 Roadmaps.
- Beginner user should see only one beginner roadmap card.
- Basic/Intermediate user should see beginner optional, intermediate recommended, and advanced locked.
- Advanced user should see only one advanced roadmap card.
- Locked card should not be selectable.
- Every roadmap card should stay within the selected career goal domain.

10. Future AI Integration

- Manual logic will be used now for predictable output and no API dependency.
- Later, AI can use the same structured payload and rules to generate more personalized roadmaps.
- AI should not break the core rules: selected career goal must control the domain, level must control available options, and Q1-Q4 remain the main inputs.
- Fallback manual logic should remain even after AI is added, so users always get a roadmap if API quota fails.

11. Implementation Notes

- Primary frontend file: public/app.js.
- Primary styling file: public/styles.css.
- Potential backend file later: src/server.js if manual generation is moved server-side.
- Contracts should be updated if output shape or generation stage changes.

12. Final Agreed Summary

Roadmap generation will be simple and goal-specific. The first four answers are enough to create a proper roadmap. Optional answers only refine the result. Manual templates will power the system now, and AI can be added later without changing the product logic.